

# Aarya Thakur

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## SUMMARY

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Final year AI DS student with hands-on experience building LLM-powered systems, agentic AI pipelines, and RAG architectures. Proven in high-pressure environments with 2 internships and 6 hackathon wins.

## EDUCATION

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**Thadomal Shahani Engineering College**

*Bachelor of Engineering in Artificial Intelligence and Data Science*

CGPA 9.43

November 2022 – June 2026

## EXPERIENCE

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**Fullstack Intern**

June 2025 – July 2025

*Augenblick Consulting*

*Remote*

- Implemented Deep Zoom Technology, improving large image rendering speed and enabling seamless web-based navigation of gigapixel images.
- Contributed to building a RAG-powered system that converts RIS files into vector embeddings, enabling accurate research paper classification, tagging, and advanced rule-based filtering.

**AI Intern**

January 2025 – April 2025

*Nimrobo AI*

*Remote*

- Created a tool enabling agentic AI to query and interpret CSV data using SQL for advanced data analysis and insights.
- Integrated citation tracking into AI Agent outputs, enhancing content credibility and reducing hallucinated outputs.
- Developed a container orchestration system that automatically spins up isolated AI agent instances with WebSocket access, and automated lifecycle management, improving scalability and reliability.

## PROJECTS

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**MatryoshkaSearch**  | *Retrieval Pipeline with Custom Embeddings* | Python, LangGraph, FastAPI, ChromaDB

- Fine-tuned embedding model on ~7K financial Q&A pairs using Matryoshka Representation Learning achieving 10–13% gains on various metrics over the base model.
- Built a LangGraph retrieval pipeline with semantic chunking-based document ingestion, query routing and cross-encoder reranking, containerized the full stack with Docker.

**CollabNext**  | *Real-Time Collaborative Editor with AI* | Next.js, TypeScript, Firebase, WebSocket, Yjs

- Built a real-time collaborative content studio with WebSockets and Yjs Conflict-free Replicated Data Types, ensuring conflict-free synchronization, and smooth multi-user editing.
- Integrated AI pipelines for ghostwriting, comic/story generation, and sentiment analysis, streamlining the content creation workflow and accelerating editing pipeline for teams.

**Metaheuristic Job Scheduler**  | *Simulated Annealing & Genetic Algorithms* | Python

- Implemented a Simulated Annealing-based solution, enabling escape from local optima and consistently outperforming greedy baseline heuristics.
- Applied Genetic Algorithm with custom chromosome encoding, crossover, and mutation strategies, achieving faster convergence and superior global exploration than greedy methods.

## ACHIEVEMENTS

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Winner of **6 Hackathons**, consistently excelling in fast-paced, high-stakes environments across diverse technical domains.

Achieved a **Perfect 10 CGPA** in the 6th Semester.

## TECHNICAL SKILLS

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**Languages:** Python, C++, TypeScript, SQL

**Databases:** PostgreSQL, MySQL, Firebase, Vector Databases

**Frameworks/Libraries:** Next.js, Django, Node.js, Express.js, Tailwind CSS, NumPy, Pandas, PyTorch, LangGraph

**Tools/Platforms:** Git, GitHub, Docker, Linux, Vercel